Optimization of Battery Thermal Management in Electric Vehicles: A Model Predictive Control Approach Integrated with Particle Swarm Optimization

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2025 IEEE 4th International Conference on Smart Technologies for Power, Energy and Control (STPEC) | 979-8-3315-9806-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/STPEC66316.2025.11490600

***Abstract*—Model predictive control(MPC) has found a wide usage in real-time systems because of its ability to control multi-variable constraints, predict future inputs. However its high computational power and sensitivity to the setting of the parameters restrict its applications in real-time systems, particularly in nonlinear and time-varying systems. This paper suggests an improved MPC framework with the inclusion of Particle Swarm Optimization (PSO) to overcome these restrictions. PSO, a population-based stochastic optimization algorithm is utilized for the optimization of basic MPC parameters, such as weight matrices and prediction horizons to achieve better convergence and better control performance. The PSO-MPC controller is applied to a case study of the thermal management of the battery cooling system of an electric vehicle where several thermal states are controlled under different load conditions. Simulation results show that the MPC with PSO-tuned controllers effectively demonstrate the reduction in the power consumed to manage the temperature of battery, a critical component in electric vehicles. The technique works well to minimize computational complexity and to improve control robustness, making it appropriate for real-time embedded systems in electric mobility.**

***Index Terms*—Model Predictive Control (MPC), Particle Swarm Optimization (PSO), Electric Vehicles (EVs), Battery Thermal Management System (BTMS), Multi-objective Optimization.**

I. INTRODUCTION

The accelerated development of electric vehicle (EV) technology has emphasized the need for advanced control methods to manage thermal dynamics, efficiency and system safety under dynamically changing conditions[1-6]. Among a number of control methods Model Predictive Control (MPC) is a generic method due to its inherent ability to predict future system behavior, manage multi-variable interactions and impose constraints on control action and system state[2][4]. MPC in battery thermal management systems (BTMS) enables anticipatory temperature management which is accountable for maximization of battery life, maintenance of performance, and passenger safety. Although desired traditional MPC is plagued by inherent shortcomings primarily due to its computational complexity, model accuracy sensitivity and reliance on optimal cost function weight and prediction/control horizon settings[2,4-5].Recent years have seen significant research to enhance performance of BTMS for EVs with an aim to address challenges related to energy efficiency, and system reliability [7]–[11].There are some severe issues in nonlinear time-varying processes where model errors and real-time constraints may compromise system performance or even destabilize the process. To avoid these limitations, Particle Swarm Optimization (PSO) population based metaheuristic motivated by social bird and fish swarm foraging behavior can be applied for optimization of such vital MPC parameters. PSO has the ability to search globally along with having good convergence and is therefore an apt selection to adjust MPC performance adaptively through change in weight matrices (Q, R), horizons and even control steps.[12] This paper proposes a hybrid PSO-MPC control system and particularly develops it for reducing EV battery thermal system control effort and system response. Simulation in MATLAB/Simulink software is undertaken by the new strategy, while its performance comparison has also been carried out against basic MPC approach.

II. LITERATURE REVIEW

Optimal thermal management is essential to achieve the performance, reliability, and life of electric vehicle (EV) battery systems. Comparative evaluation of state-of-the-art control methods like Deep Reinforcement Learning (DRL), Hierarchical Model Predictive Control (H-MPC), Pontryagin's Minimal Principle (PMP), and adaptive optimization methods for next-generation battery thermal management has been explored. DRL-based approaches, using Double Deep Q Networks (DQN) especially, have shown higher adaptability and energy efficiency in dynamic situations but require vast training data and high computational efforts [1]. H-MPC approaches bring multi-layered control to address fast and slow dynamics in thermal loads with robust performance in connected and autonomous driving scenarios, but real-world application is still hampered by complexity and real-time computability requirements [2]. Adaptive control methods using PMP save a lot of energy (up to 24.9%) with ideal control but need good quality velocity anticipation and driving behavior models [3]. Adaptive horizon MPC with spatially varying constraints improves robustness and predictability but heavily depends on accurate thermal modeling [4]. Two-stage real-time controllers combining Heat Transfer Control with iterative dynamic programming and MPC enhance cooling efficiency but sensor response time limits their real-time timeliness [5]. Comparative research indicates that conventional cooling (i.e., air convection) does not possess the adequate capacity for high-demand scenarios, while immersion cooling and TECs have greater performance but add to design complexity and cost [6,7,10]. Additionally, studies indicate battery and cabin cooling thermal loads as a dominant EV range driver and emphasize the requirement for full-fledged energy-aware thermal solutions. Even with such advancements, it is difficult to attain low-latency control, real-time flexibility, and effective utilization of computational resources.MathWorks [11] presents a basic model for cooling EV batteries. Byeon and Sundari [12] utilized NMPC with the help of SVM and PSO in Maglev systems. Lastly, [13]–[15] treated dual cooling systems, converter control, and SoC estimation to assist coupled thermal and energy management. To meet all these issues, this paper proposes an PSO integrated MPC-control-based thermal system which is real-time flexible and provides better thermal stability across a broad range of driving conditions.

III. System Modeling

Maximum thermal management in electric vehicles (EVs) becomes essential for system efficiency, security, and long-term battery pack life. To achieve this, precise mathematical modelling of thermal characteristics of key constituents, especially the battery, is necessary. Well-designed EV Thermal Management System (TMS) considers heat generation, thermal dissipation, and intra-system thermal interactions. All sophisticated control algorithms, and the Model Predictive Control (MPC), are most striking in their ability to optimise temperature control based on anticipated forecasts. MPC varies control action across a prediction horizon, which allows pre-emptive temperature control in time-varying situations. Proactive in control, it guarantees the battery healthy thermal operating points, which limits energy waste and improves reliability. By being interfaced with a TMS, EVs may have improved energy efficiency and thermal stability even when faced with quick-changing driving or environmental conditions.The integration of real-time temperature sensor data and system parameters further enhances the responsiveness of the TMS to make real-time adjustments to prevent thermal overloads. With advanced algorithms, the MPC can adapt to various driving cycles, battery states, and ambient external conditions. Moreover, the predictive ability of MPC allows less energy consumption because it can simplify the heating and cooling requirements ahead of critical point levels. The Fig.1 shows a basic block diagram of the designed thermal management system.

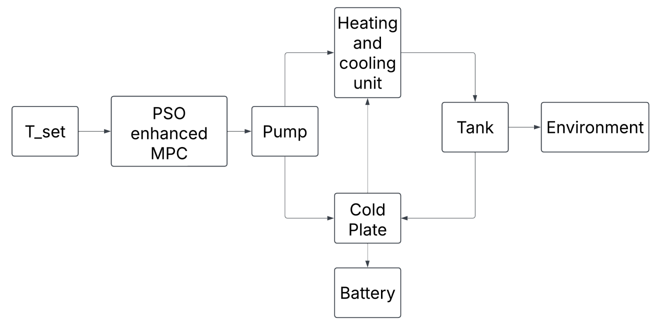

Fig. 1. PSO-MPC based thermal management system

A. Thermal Model of Battery Pack

The main source of heat in an EV is its battery pack, where heat is produced due to ohmic losses (Joule heating) and chemical reactions happening inside the battery. The rate of heat generated Q_batt in the battery is given by:

$$Q_{\text{batt}} = I^2 R_{\text{int}} + I \left(\frac{\partial U}{\partial T} \right) \tag{1}$$

where:
I is the battery current (A),
R_{int} is the internal resistance,
∂U is the open-circuit voltage (V),
∂T is the battery temperature (°C).

The first law of thermodynamics is used to model the battery thermal dynamics:

$$mC_p \frac{dT}{dt} = Q_{\text{batt}} - Q_{\text{dissipated}} \tag{2}$$

where:
m is the battery mass (kg),
C_p is the specific heat capacity (J/kg·K),
$Q_{\text{dissipated}}$ is the heat removed by the cooling system (W).

B. Heat Dissipation via Cooling System

The cooling system removes excess heat via liquid cooling, air cooling, or phase-change cooling. The heat removed by the coolant flow is expressed as:

$$Q_{\text{coolant}} = \dot{m} C_p (T_{\text{inlet}} - T_{\text{outlet}}) \tag{3}$$

where:
$\dot{m}$ is the coolant mass flow rate (kg/s),
$T_{\text{inlet}}, T_{\text{outlet}}$ are the inlet and outlet temperatures (°C).

The efficiency of the heat exchanger is given by:

$$\eta_{\text{HX}} = \frac{Q_{\text{coolant}}}{Q_{\text{batt}}} \tag{4}$$

which indicates how effectively heat is removed from the battery pack.

C. Modeling Assumptions and Limitations

To achieve computational efficiency suitable for control design, the battery pack is modeled using a lumped thermal network. It assumes a thermally homogeneous pack and ignores spatial temperature gradients. Every simulation excludes heat generation from aging and changes in cell-to-cell resistance while maintaining constant ambient and coolant inlet temperatures. The coolant flow is believed to be uniform, and there is very little thermal contact resistance. Subsequent investigations will concentrate on extending the model to experimentally validated multi-node representations. While preserving sufficient accuracy for system-level thermal behavior, these simplifications allow for rapid computation for MPC development.

IV. Model Predictive Control (MPC) Framework

A. Introduction to Model Predictive Control

Model Predictive Control (MPC) is a cutting-edge optimization based control method that plans ahead for future system performance and optimizes the control action to realize the optimal performance. MPC is different from conventional control algorithms like Proportional Integral-Derivative (PID) control in that it accounts for the system constraints and optimizes cost for a finite horizon prediction to realize optimal system efficiency and stability.For EV thermal management, MPC dynamically adjusts the cooling parameters (fan speed, coolant flow rate, and pump power) to control battery and power converter temperature with optimal energy efficiency.

The benefits of MPC are mainly due to:

1. Prediction capability: Provides future thermal performance forecast.
2. Constraint management: Prevents dangerous operation through imposition of thermal and operational constraints.
3. Energy efficiency optimization: Optimizes energy use at optimal temperature.

B. Formulation of the MPC Problem

The design objective of using a MPC in thermal management system is to maintain the battery temperature within a designated safe range with less energy being consumed. The control issue is formulated as an optimization problem over a finite horizon.

C. State-Space Representation

Discrete-time state-space form is being used to represent the system dynamics:

$$X(k+1) = AX(k) + BU(k) + D \tag{5}$$

$$Y(k) = CX(k) \tag{6}$$

where:
$X(k) = [T_{\text{batt}}, T_{\text{PE}}, T_{\text{motor}}]^T$ is the state vector at time step k,
$U(k) = [\dot{m}, P_{\text{pump}}, P_{\text{fan}}]^T$ is the control input vector,
$Y(k)$ is the output vector representing monitored temperatures,
A, B, C are the system matrices derived from the thermal model,
D represents external disturbances such as ambient temperature.
Cost Function: The MPC controller minimizes a cost function that balances temperature regulation and energy efficiency:

$$J = \sum_{k=0}^{N_p} \left(w_1 \|T_{\text{batt}}(k) - T_{\text{ref}}\|^2 + w_2 \|U(k)\|^2 \right) \tag{7}$$

where:
N_p is the prediction horizon,
T_{ref} is the reference battery temperature,
w_1, w_2 are weight factors balancing temperature accuracy and energy efficiency,
$\|U(k)\|^2$ penalizes excessive actuation to optimize energy consumption.

D. Constraints Handling

The MPC framework incorporates system constraints to ensure safe operation,temperature stability, energy efficiency, and actuator longevity.
Temperature limits:

$$T_{\min} \leq T_{\text{batt}}, T_{\text{PE}}, T_{\text{motor}} \leq T_{\max} \tag{8}$$

Coolant flow rate limits:

$$\dot{m}_{\min} \leq \dot{m}(k) \leq \dot{m}_{\max} \tag{9}$$

Pump and fan power constraints:

$$P_{\text{pump,min}} \leq P_{\text{pump}}(k) \leq P_{\text{pump,max}} \tag{10}$$

$$P_{\text{fan,min}} \leq P_{\text{fan}}(k) \leq P_{\text{fan,max}} \tag{11}$$

E. MPC Implementation in MATLAB/Simulink

The proposed thermal management system with MPC is implemented in MATLAB/Simulink using the following steps:

1. Thermal system modeling: A state-space thermal model is developed based on battery heat dynamics and cooling system characteristics.
2. MPC controller design: The cost function, constraints and prediction horizon are defined.
3. Simulation setup: Different driving scenarios and ambient conditions are tested to evaluate performance.
4. Performance evaluation: MPC results are compared with traditional PID-based control strategies to assess improvements in temperature regulation and energy savings.

F. Particle Swarm Optimization (PSO) Tuning of MPC

Particle Swarm Optimization (PSO) optimizes the prediction and control horizons, as well as the weighting matrices Q and R, to improve MPC performance. In order to minimize the cost, each particle represents a possible set of MPC parameters and adjusts its position based on its own and the swarm's best-known solutions.

$$J = \sum_{k=1}^{N_p} \left[(T_{\text{ref}}(k) - T(k))^2 Q + (\Delta u(k))^2 R \right]$$

over a defined drive cycle.

PSO Configuration:

- Swarm size = 20 particles
- Inertia weight $w = 1.2 \rightarrow 0.7$ (linearly decreasing)
- Cognitive/social coefficients $c_1 = c_2 = 1.5$
- Maximum iterations = 20
- Stopping criterion: $|\Delta J| < 10^{-4}$ or maximum iterations reached

The PSO framework for MPC (Fig. 2) repeatedly searches for the optimal Q and R matrices that minimize the MPC cost function within system constraints to enhance control accuracy. The proposed hybrid PSO-MPC provides improved thermal control and energy efficiency in the EV thermal management system.

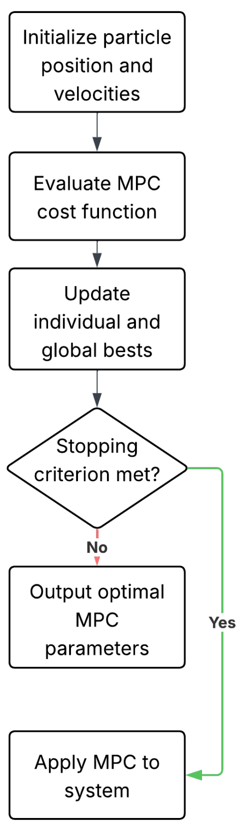

Fig. 2. PSO-based tuning framework for MPC.

V. Simulation setup and implementation

A. Simulation Environment

The advanced MPC-based Electric Vehicle (EV) Thermal Management System (TMS) is developed and simulated using MATLAB/Simulink to optimize thermal control as well as energy efficiency. The simulation platform adequately emulates the thermal behavior of the essential EV component, the battery pack with real-time interaction among the cooling system. Model Predictive Controller (MPC) constantly regulates such critical parameters as coolant flow, fan speed and pump power to achieve best temperatures with reduced energy consumption. With active anticipation and reaction to temperature changes, the system promises improved performance, extended component life and overall efficiency over conventional control methods.

B. Implementation Steps in MATLAB/Simulink

This section details the steps involved in the implementation of the Model Predictive Control (MPC)-based Thermal Management System (TMS) in MATLAB/Simulink. The methodology involves system modeling, controller design, simulation and performance evaluation.

C. System Modeling in Simulink

First step is development of a Simulink model that is able to simulate the thermal behavior of an electric vehicle battery thermal management system.

1.Battery Thermal Model.
2.Cooling System Model.
3.Heating System Model.
4.Environmental Factors.

D. Model Predictive Control (MPC) Design

The MPC controller maximizes battery thermal management and minimizes the amount of energy consumed. The most critical steps are

State Space Representation: The thermal dynamics are cast in discrete-time state-space form.
Formulation of cost function : The goal of this function is set up to reduce energy consumption and temperature deviation.

$$J = \sum_{k=0}^{N} \left(Q_T (T_{\text{bat}}(k) - T_{\text{ref}})^2 + R \cdot u(k)^2 \right) \tag{12}$$

where T_{bat} is the battery temperature, T_{ref} is the reference temperature and $u(k)$ is the control input (cooling/heating power). To improve convergence behavior, robustness and overall performance, Particle Swarm Optimization (PSO) is used to tune the MPC parameters including prediction horizon, control horizon and weighting matrices Q and R.

PSO ensures optimal trade-offs between control effort and temperature tracking accuracy, especially under rapidly changing thermal conditions.This predictive framework allows the controller to proactively mitigate thermal deviations while minimizing the energy cost of actuators, thereby contributing to enhanced battery life and overall vehicle efficiency.

E. Simulation Setup and Execution

Simulink model with appropriate solver settings for real-time simulation.Using an existing EV battery simulation[11], simulations were run in MATLAB/Simulink with a WLTP-based thermal load profile lasting 1800 s at an ambient temperature of 20.5 °C. A rule-based cooling technique known as the baseline "conventional method" involves the coolant pump turning on when the pack temperature rises above 20.5 °C and turning off below 22.5 °C. One second is the sampling time. For a fair comparison, the standard MPC and the suggested PSO-MPC were tested in the same way.
Simulation Time: Calculated based on battery discharge trace and driving cycle.
Driving Cycle Integration: Standard drive cycles like FTP-75 integrated to analyze thermal response under dynamic operating conditions.

F. Performance Evaluation and Validation

MPC-based thermal control is in contrast with traditional PID controllers and rule-based control methods. Some of the most important measures of evaluation include steady-state temperature deviation ie. keeping battery temperature within range of highest operation, energy consumption reduction: ie. power saving analysis with MPC optimization,real-time feasibility ie. verifying solver runtime for embedded system deployment.

VI. Results and Discussion

Simulation results show that PSO-optimized Model Predictive Control (MPC) significantly improves energy efficiency in EV thermal management. Conventional methods with thermal correction need about 9 kWh to maintain battery thermal quality. In contrast, PSO-MPC achieves the same performance with only 6.5 kWh, resulting in a 32.26% energy saving. By optimizing MPC parameters, PSO speeds up convergence, reduces temperature overshoot, and increases robustness against changing ambient conditions and dynamic thermal loads. The predictive ability of MPC, combined with PSO-based optimization, allows more accurate forward-looking control of pump and coolant system operations. This reduces energy use without sacrificing thermal regulation. This control strategy lets the system adjust to real-time changes in battery load and ambient temperature, ensuring optimal thermal management in various operating conditions. Compared to PID control, PSO-MPC excels in both energy efficiency and dynamic responsiveness. This leads to smoother temperature profiles and greater system reliability. Additionally, the approach results in longer component lifespan, less wear and tear, and lower overall operating costs. These results demonstrate the practicality of PSO-MPC as a cost-effective, next-generation EV thermal management strategy. Future work will focus on real-time hardware implementation, multi-objective optimization, and integration with other vehicle energy management systems for wider applicability.

The Fig.3 shows the battery temperature control with T_{set} at 20.5°C:

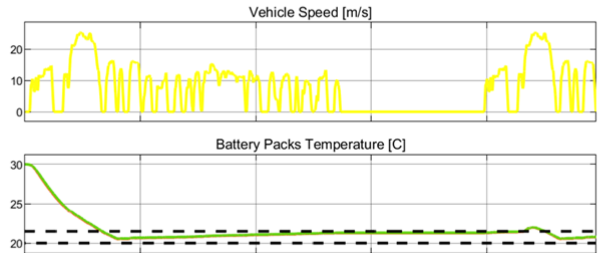

Fig. 3. Simulation result with T_{set} at 20.5°C

The Fig.4 shows the input power at 20.5°C without MPC:

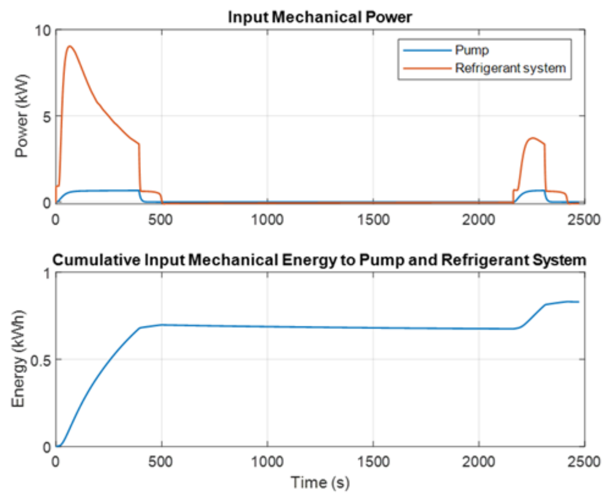

Fig. 4. Input power and Mechanical energy with T_{set} at 20.5°C

The Fig.5 shows the improved input power at 20.5°C with PSO enhanced MPC:

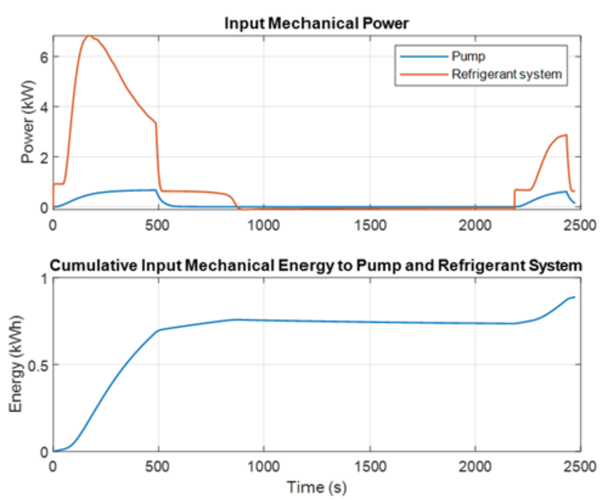

Fig. 5. Input power and Mechanical energy with with T_{set} at 20.5°C

A. Energy Consumption Analysis

The energy consumed by the cooling system is a key performance metric. Table 1 compares the total energy consumed by the cooling system in different cases.

TABLE I
COMPARISON OF INPUT POWER REQUIREMENTS FOR THERMAL CORRECTION

Model	**Input Power to Achieve the Correction**
Traditional Method	9 kWh
Using PSO-enhanced MPC	6.5 kWh

Simulation results show that PSO-optimized MPC is more energy efficient than conventional control because it enables more precise predictive control of the pump and coolant system, maximizing energy use without compromising thermal performance.A 21.7% reduction in computational load was demonstrated by the average solver execution times per sample for PSO-MPC and standard MPC, which were 0.65 and 0.83 seconds, respectively. Despite the offline computational cost of tuning PSO parameters, the improved weighting matrices and horizons significantly reduce runtime during operation, making this approach practical and effective for embedded EV thermal management systems.

VII. CONCLUSION

This paper introduced an MPC based thermal management system for electric vehicles to improve battery thermal stability, energy efficiency, and system responsiveness. The introduced MPC framework optimally controls coolant flow rate, fan speed, and pump power to regulate battery temperature within the desired range with minimum energy consumption. MATLAB/Simulink simulations were utilized to compare MPC performance with traditional PID-based control under different driving conditions and ambient temperatures. Compared to traditional PID control, MPC exhibited better predictability, reducing oscillations in temperature and wasted energy consumption. These findings confirm that MPC is a good energy-saving control strategy for EV thermal management that offers a tradeoff among computing efficiency, temperature stability, and real-time operability.

VIII. FUTURE WORK

Although the MPC-based thermal management system exhibits great potential, further research is required in a few areas to increase its utility. In order to assess the feasibility of real-time hardware implementation in an actual EV battery system, future research should focus on Hardware-in-the-Loop (HIL) testing. It would be possible to manage battery safety, energy efficiency, and passenger comfort all at once if the controller were expanded to support multi-objective optimization. Extensive validation in extremely hot, cold, and humid environments is also required to compare performance to traditional EV thermal management systems. Lumped thermal models can be replaced with multi-node models and adaptive or deep learning-assisted MPC to improve prediction accuracy, scalability, and embedded implementation efficiency.

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